DIGITAL TWIN CHALLENGE 2026

Reimagining and Reforming Healthcare in India Summit 2026

HAPPIEST HEALTH

#ReimaginingHealthcare

GLUCOSE DIGITAL TWIN

Fusing health records and wearable streams to warn of high and low glucose two hours ahead

A type 2 diabetes proof of concept. All data synthetic.

AG

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VS

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Team “TeamTwins”

Why this matters

101 M — people with diabetes in India

136 M — people with prediabetes

5 min — CGM reading interval

2 h — warning time targeted

800 — synthetic patients, 14 days

ICMR INDIAB 2023 for prevalence. South Asians develop type 2 diabetes at lower BMI and younger ages.

Where care breaks down

Event: 15 min out of range

High above 180, low below 70 mg/dL

Horizon: up to 2 hours ahead

Seen rarely

A patient meets a clinician a few times a year, so a high above 180 or a low below 70 between visits is never seen.

Data unlinked

CGMs and smartwatches stream every few minutes, but the streams are not joined to labs, diagnoses, medicines or genetics.

Too much to read

No doctor can scan every trace, so risk is found after the event rather than before it.

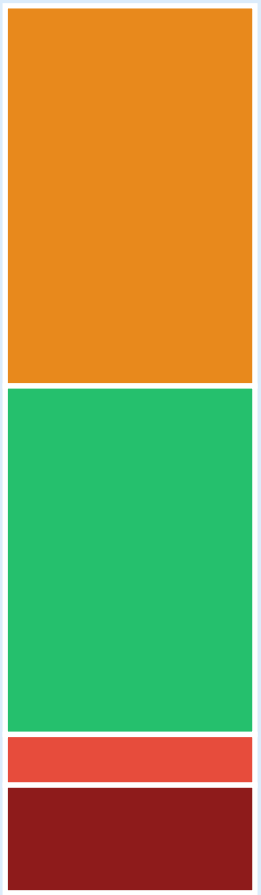
No priority order

Without ranking, a clinic cannot tell which patient needs attention first.

The question we answer

Can a patient's record (labs, diagnoses, medicines, genetics) fused with wearable streams warn of a glucose event above 180 or below 70 mg/dL up to two hours ahead, explain why, and rank the whole panel by who needs attention first?

Glucose zones (mg/dL)



Level: high >180

In range 70 to 180

Level 1 low <70

Level 2 low <54

Thresholds follow the international consensus on CGM metrics: time in range 70 to 180 mg/dL, with level 1 low below 70 and level 2 low below 54.

How we define an event

Out of range for 15 min

an event is at least 3 consecutive 5 minute readings above 180 or below 70, so single sample sensor spikes are ignored

Onset after an hour in range

an onset counts only after one hour fully in range, so we predict new events, not ones already underway

Look ahead 5 to 120 min

the label looks only at t+5 to t+120 minutes; features at t use data up to t

Alert needs confirmation

risk must exceed the threshold now and 10 minutes earlier (2 readings)

Why these choices matter clinically

- **15 min** — sustained excursion, not noise
- **2 h** — time to act: snack, dose, call
- **3 levels** — sensitive, balanced, precise
- **1 h** — in range before an onset counts

Hypoglycaemia is rare (2.7% of in range moments), so every precision figure is read against that base rate.

The organiser: Happiest Health

An integrated wellness enterprise based in Koramangala, Bengaluru (Happiest Health Systems Pvt. Ltd.), announced in April 2022 and promoted by Ashok Soota. Mission: inspire everyone to embrace wellness as a way of life.

Knowledge	Diagnostics	Healthcare	Wellness
Magazine, daily Healthzine, podcasts, videos, summits	Routine and specialised blood tests, Bengaluru	Dental and orthopaedics now; ENT, eye, paediatrics, obesity, urology planned	Ayurveda, mental health, physio; Wellness Catalyst for corporates

Ashok Soota, promoter

- **Wipro** — grew its IT business into India's largest computer company
- **2007** — founding Chairman, MindTree (IPO)
- **2020** — founding Chairman, Happiest Minds (IPO)
- **2021** — founded SKAN: ageing and neurological research

Ashirvadam trust (2011). Forbes Asia Top 15 philanthropists, 2022.

The challenge: Digital Twin Challenge 2026

- **~~18 Sep~~** — Submissions open
- **~~24 Oct~~** — Code due, 7 PM IST
- **~~Date TBA~~** — Top 10 jury presentation (virtual)
- **~~20 Nov~~** — Top 5 showcase, Bengaluru

Prizes: Rs 1,00,000 winner, Rs 50,000 first runner up, Rs 30,000 second runner up, two Rs 10,000 recognition awards, plus internship (PPI) opportunities. Travel and stay for two members of each finalist team are sponsored.

Why our project fits

The summit programme centres on evidence based care, continuous health modelling and the opportunities and limits of AI. Our twin is exactly a continuous health model, with honest limits stated.

Sources: happiesthealth.com/about-us, district.in event listing, Unstop challenge page.

Challenge brief and where each point is met

Requirement	How this submission meets it	Where
Static data: simulated EHR with demographics, diagnoses, labs, genetic markers	Synthea plus generated labs, medicines, 5 SNPs and a polygenic score	ehr/build_ehr.py
Dynamic data: wearable or CGM time series (HRV, glucose, sleep, steps)	Dexcom style CGM, heart rate, HRV, steps, sleep stages, SpO2, wrist temperature, insulin and carb logs	wearables/
Fuse both streams	5 minute timeline; EHR joined only from labs dated before each timestamp	ingestion/align.py
Predict an adverse event	P(glucose above 180) and P(below 70) within 2 h, plus 1 h and 2 h forecasts	models/events.py
Conceptual doctor dashboard	Working Streamlit app: triage, patient view, live monitor, results	dashboard/app.py
Synthetic or open data only	No real patient data; MIMIC IV not used	CONTRACT.md
Architecture diagram and presentation (PDF or PPT)	Generated from code so numbers stay in sync	docs/
Open source licence details	MIT licence; Synthea is Apache 2.0	LICENSE
Demo video	Unlisted YouTube link is in the README	README

Submission checklist

- ✓ Public GitHub repository
- ✓ README: team, title, problem, stack, license
- ✓ Open source license (MIT)
- ✓ Architecture diagram, PDF
- ✓ Presentation, PDF
- ✓ Synthetic data only
- ✓ Demo video on YouTube (link in README)
- ✓ Folder name TeamTwins_IGDTUW (stated in README)

All submission items are ready.

Synthetic EHR (static stream)

patients.csv

age, sex, BMI, HbA1c, fasting glucose and insulin, HOMA IR, lipids, eGFR, ALT, status, smoking, family history

conditions.csv

T2D, prediabetes, hypertension, dyslipidemia, obesity, CKD 1 to 3, retinopathy, neuropathy, CAD

labs.csv

4 to 8 visits in the 730 days before 2026-09-30; 10 lab tests; eGFR by CKD EPI 2021

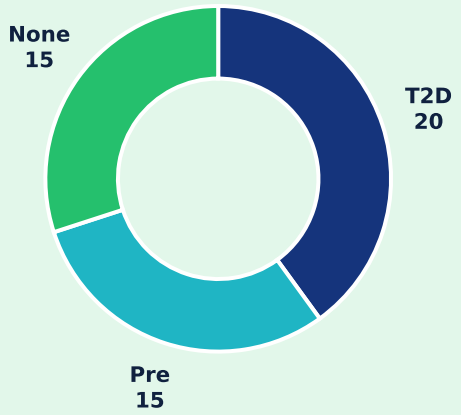
medications.csv

metformin, insulin basal and bolus, SGLT2, DPP 4, sulfonylurea, statin, ACE or ARB

genomics.csv

5 SNPs, risk allele counts, polygenic score; labelled SIMULATED

Cohort mix



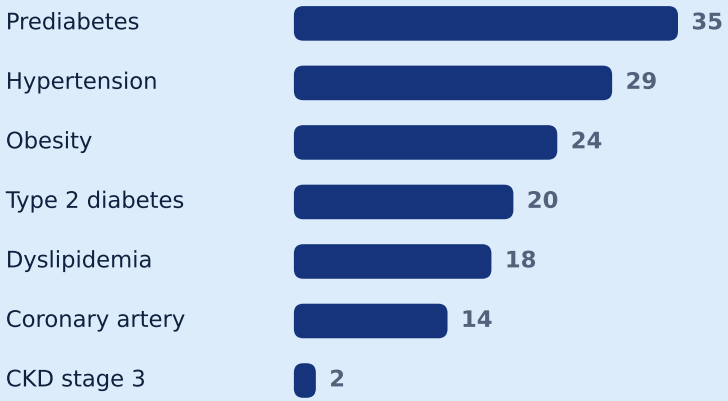
Full profile: 800 patients

- 38% type 2 diabetes
- 31% prediabetes
- 31% no diabetes

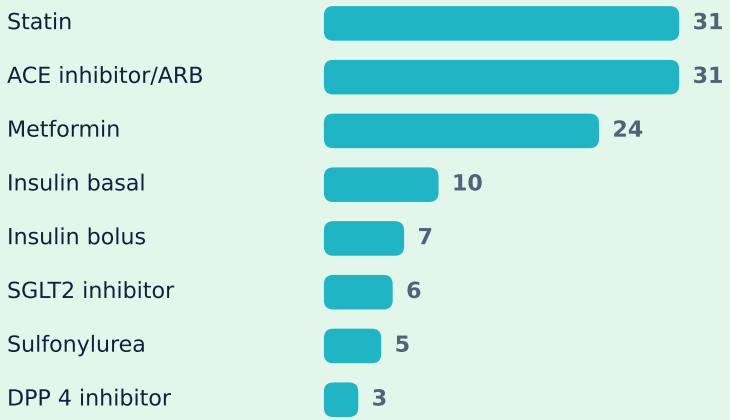
Sample: mean age 58.7 (37 to 78), BMI 30.0, HbA1c 6.6; 27 F, 23 M.

Sample profile: 50 patients (T2D, prediabetes, none)

Conditions (sample, n=50)



Medicines (sample, patients)



Genetic markers (simulated)

SNP	Gene	RAF	OR
rs7903146	TCF7L2	0.30	1.40
rs5219	KCNJ11	0.36	1.14
rs1801282	PPARG	0.85	1.14
rs9939609	FTO	0.40	1.20
rs13266634	SLC30A8	0.65	1.12

Wearable streams at native rates

Device (simulated)	Metric	Rate and realism
Dexcom G7 style CGM	glucose	5 min, AR(1) noise, 30 min warm up, day 10 sensor change, 15 to 120 min dropouts
Apple Watch style	heart rate	1 min near activity, else 5 to 10 min
	HRV, resting HR	3 to 5 day spot reads, 4 to 8 overnight; 1 resting a day
	steps, energy	irregular segments up to 26 min
	sleep stage	in bed, awake, core, deep, REM
	SpO2, wrist temp	overnight; 1 per night
Manual log, insulin pen	carbs, bolus, basal	meals logged with p = 0.35 to 0.95; boluses 0.5 to 0.97; about 25% carb error

EHR to physiology link

HOMA IR, HbA1c, smoking, polygenic score, age	→ insulin sensitivity falls
BMI	→ glucose clearance falls
eGFR	→ insulin clears slower
HbA1c, years since dx, DPP 4	→ beta cell response
SGLT2 inhibitor	→ lower glucose peaks
Sulfonylurea	→ higher hypo risk
Metformin	→ hepatic output minus 8%

Built to be messy, like real devices

- **15-120** min CGM dropouts
- **0.35-0.95** meal log probability
- **~25%** carb mis estimate
- **1 min** truth, never an input

Pipeline: five stages, each with a test

1 Generate
Synthea EHR, labs, medicines, SNPs; EHR conditioned CGM and watch simulation

2 Align
raw streams to 5 min grid; interpolate gaps of 15 min or less; keep gap flags

3 Fuse
join EHR per day from labs, conditions, medicines dated before that day

4 Predict
LightGBM classifiers and quantile forecasters; patient grouped 5 fold CV

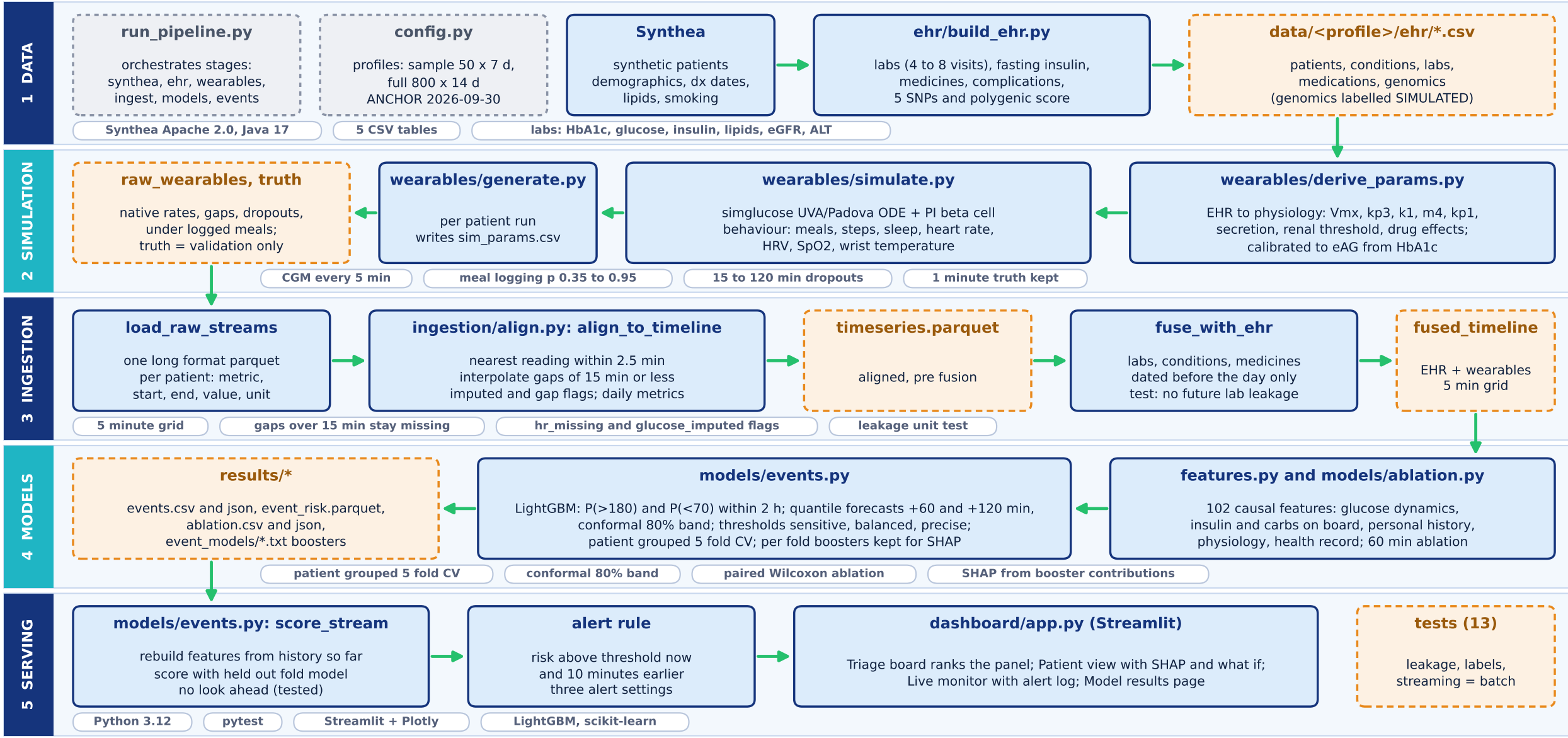
5 Explain, triage
SHAP, what if, alert log, ranked panel in Streamlit

Stage, module and key setting

Stage	Module	Key setting
Generate	ehr/build_ehr.py, wearables/simulate.py	800 patients x 14 days (full profile)
Align	ingestion/align.py	5 min grid; gaps of 15 min or less filled
Fuse	ingestion/align.py: fuse_with_ehr	labs dated before the day only
Predict	models/events.py, features.py	500 trees, 15 leaves, patient grouped CV
Explain	dashboard/app.py	triage, patient, live monitor, results

Integrity controls

- ✓ **Lab join** — only labs dated before each day
- ✓ **Label window** — t+5 to t+120 min only
- ✓ **Patient grouped CV** — no patient in train and test
- ✓ **Streaming = batch** — score_stream equals out of fold
- ✓ **Hidden truth** — 1 min glucose never a feature
- ✓ **Disclosed bias** — thresholds on pooled scores



Blue = code module Cream dashed = data artifact Grey dashed = orchestration Green arrow = data flow

Feature groups (all strictly causal)

Glucose dynamics

lags 15/30/60 min, rates of change, acceleration, 1 h and 3 h min/max/mean/SD, smoothed slope, minutes to reach 70 or 180

Insulin and carbs

insulin on board (4 h half life), carbs on board (1 h), ratio, minutes since meal and bolus

Personal history

lows and highs over 24 h and 72 h, CV, time below 90 or above 160, time since last low, same hour previous 3 days

Physiology

HR excess, steps, energy, sleep, resting HR trend, SpO2 minimum, wrist temp, time of day, gap flags

Health record

age, BMI, HbA1c, lipids, eGFR, ALT, 2 year slopes, HOMA IR, comorbidities, medicine flags, SNPs, smoking, family history

Models and training

Component	Specification
Classifiers	LightGBM binary; 500 trees, lr 0.03, 15 leaves, min 300 per leaf, subsample 0.8, colsample 0.5, L2 10
Forecasters	LightGBM quantile (10, 50, 90%) on change in glucose at +60 and +120 min
Forecast band	conformalised to 80% coverage on held out patients
Baselines	trend rule, wearable only model, persistence
Alert rule	above threshold now and 10 min earlier
Explanations	per fold boosters kept for SHAP

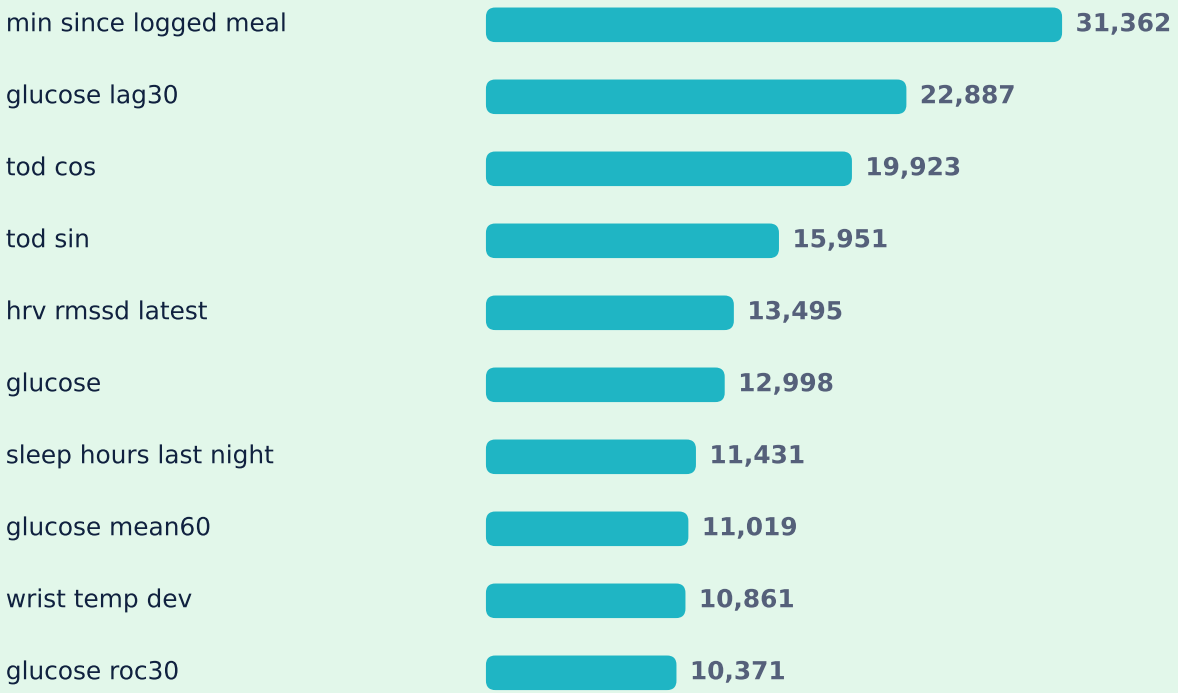
Why heavily regularised

0.03 **fused AP lost before tuning**

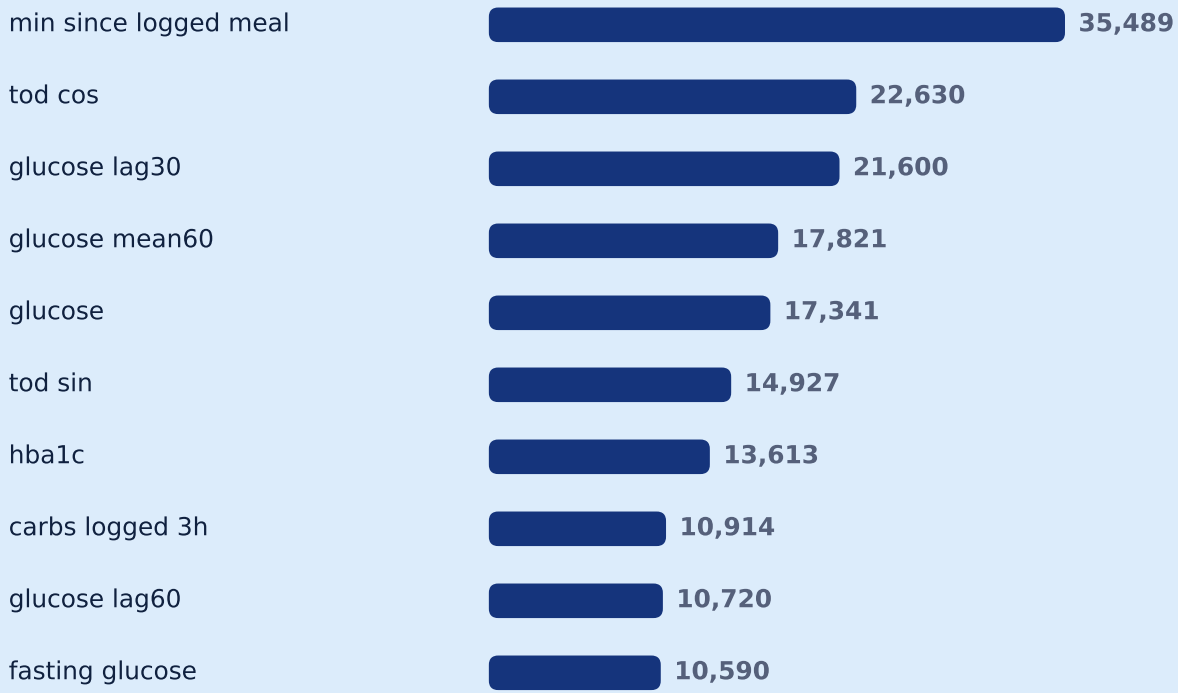
EHR columns are constant per patient, so looser trees memorise training patients. Shallow trees, 300 sample leaves and L2 10 fixed that on held out patients.

Patient grouped 5 fold CV: no patient is in both train and test folds.

Top features: wearable only model



Top features: fully fused model



Reading the importance charts

102 **input features in the event model**

Minutes since the last logged meal, recent glucose and time of day lead for both models. In the fused model HbA1c and fasting glucose enter the top 10, which is the record acting as a prior. Importances are LightGBM split gain for the 60 minute forecast ablation (sample profile).

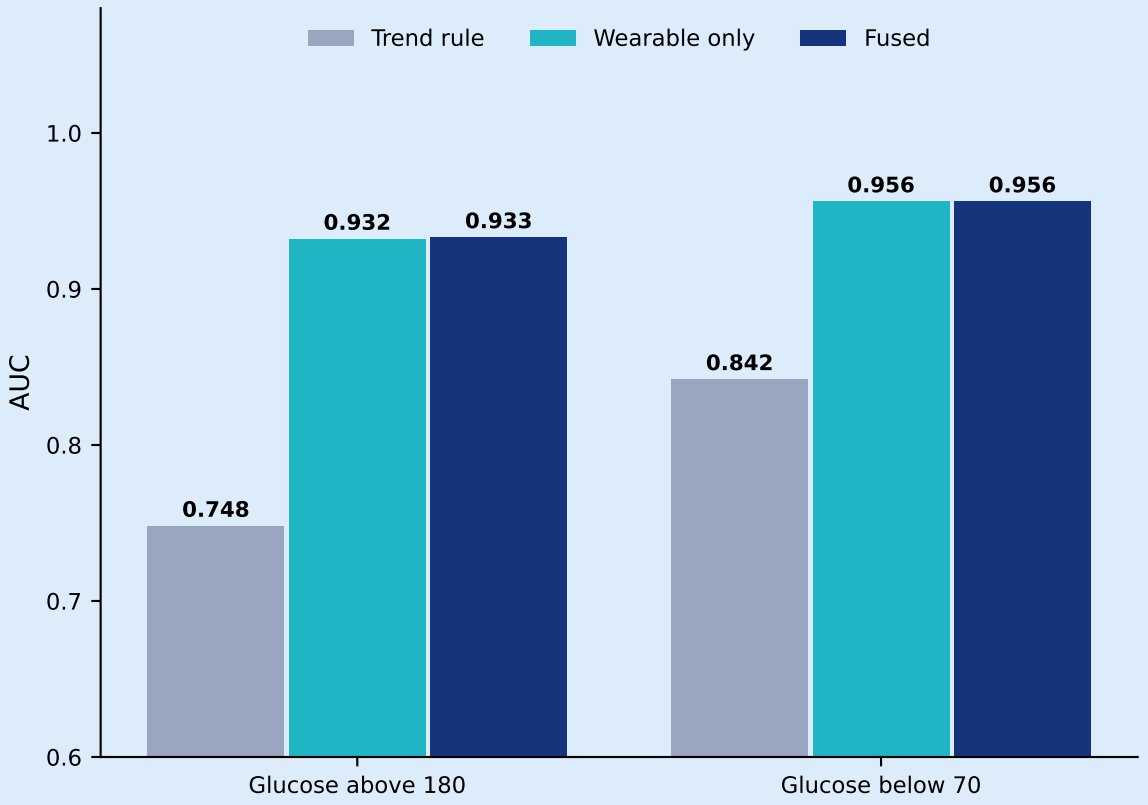
800 patients, 14 days each

5 fold CV grouped by patient

1.2 M held out test moments

2.7% hypo base rate

AUC against a trend rule



Held out results, balanced setting

Event	Model	AUC	AP	Ep. prec.	Caught	Warn
>180	Trend rule	0.748	0.31	0.30	93.7%	40 min
>180	Wearable only	0.932	0.66	0.50	88.6%	80 min
>180	**Fused**	**0.933**	**0.67**	**0.49**	**88.9%**	**80 min**
<70	Trend rule	0.842	0.16	0.19	77.0%	40 min
<70	Wearable only	0.956	0.47	0.38	70.6%	70 min
<70	**Fused**	**0.956**	**0.47**	**0.35**	**73.7%**	**70 min**

Episode precision: share of alert episodes followed by an event within 2 h.
Caught: alert in the 2 h before glucose first left range after an hour in range.
Low glucose AP is 18 times the base rate; richer features lifted it from 0.32 to 0.47 on identical labels, 0.50 now.

HEADLINE

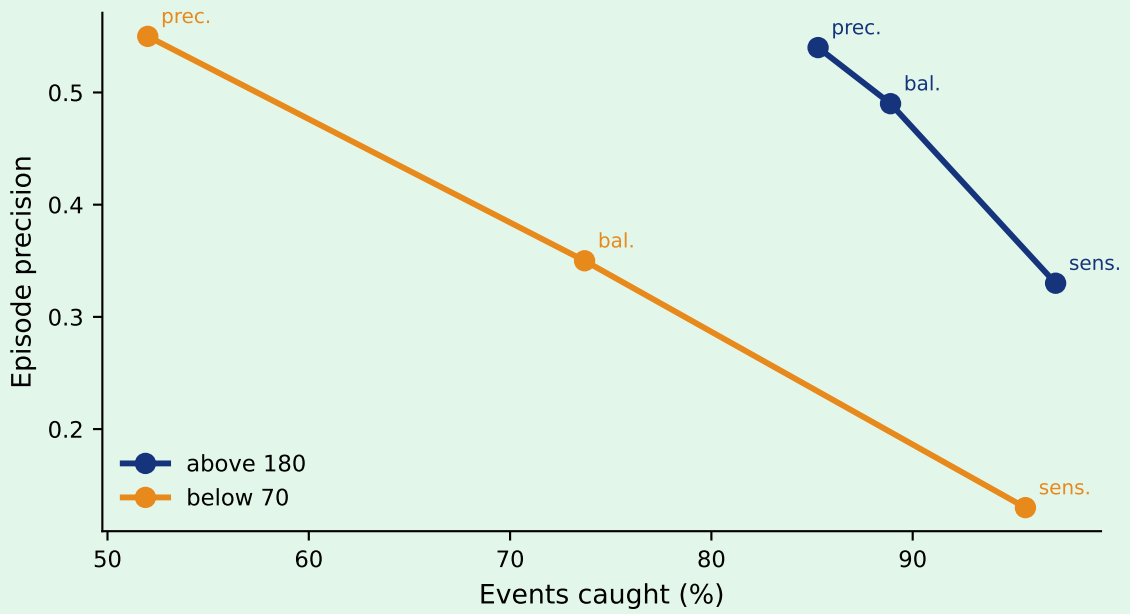
0.93 / 0.96	80 / 70 min	1.85 / 0.55
AUC high / low	median warning	alerts per patient day

Alert settings: the clinic picks the trade off

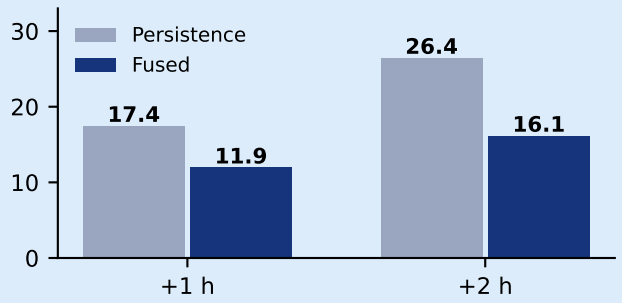
Event	Setting	Caught	Ep. precision	Row precision	Alerts / day
>180	sensitive	97.1%	0.33	0.44	2.3
>180	**balanced**	**88.9%**	**0.49**	**0.59**	**1.8**
>180	precise	85.3%	0.54	0.64	1.7
<70	sensitive	95.6%	0.13	0.21	1.3
<70	**balanced**	**73.7%**	**0.35**	**0.47**	**0.55**
<70	precise	52.0%	0.55	0.66	0.25

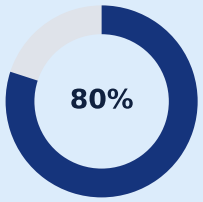
Precise low setting: 0.25 alerts a day, 55% real, misses about 48% of lows.
Sensitive: catches 96% at 13% episode precision.

Trade off curve



Forecast error, median mg/dL





80% band coverage

32% and 39% better than persistence

Reading the results

The band is conformalised on held out patients, so 80% coverage is measured, not assumed. Median error is 11.9 mg/dL at 1 h and 16.1 at 2 h, against 17.4 and 26.4 for repeating the last reading.

Does the health record help? An honest answer

After one day of CGM: no

Personal history features carry what the record knew. Fused and wearable only models tie: AUC 0.933 and 0.956.

First 24 hours: yes for highs

AUC 0.916 against 0.905, AP 0.62 against 0.59. For lows AP 0.38 against 0.40, within noise.

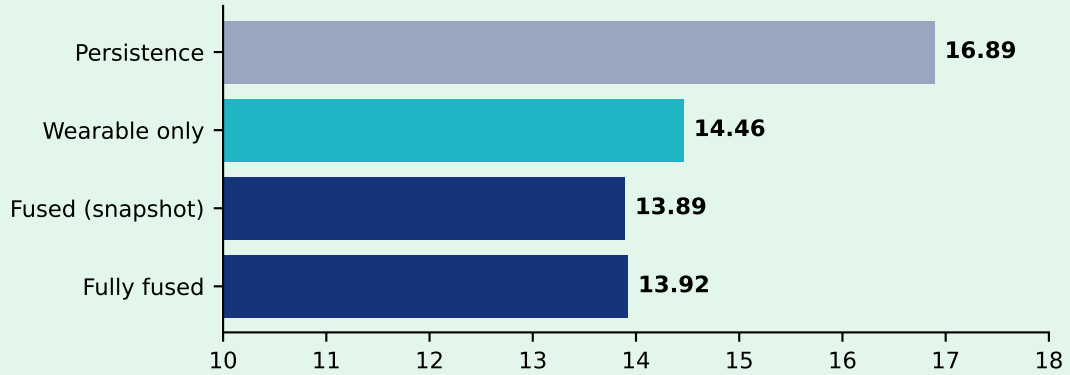
Against no history model

The record cuts 60 minute error from 13.5 to 12.8 mg/dL.

What this really shows

The simulator is driven by the EHR, so this shows the pipeline recovers that link, not that it exists in real patients.

60 minute forecast MAE, sample (50 patients)



Paired Wilcoxon tests across patients

- **0.57** mg/dL lower MAE with snapshot EHR
- **70%** of patients improve, p = 0.0002
- **p 0.57** full EHR adds nothing on top

50 patients (sample profile)

350 patient days

76,136 scored moments

326 / 41 high / low onsets

Committed results, sample profile (balanced setting)

Event	Model	AUC	AP	Ep. prec.	Caught	Warn	Alerts/day
>180	Trend rule	0.743	0.32	0.30	96.3%	45 min	4.51
<70	Trend rule	0.856	0.11	0.20	75.6%	45 min	0.65
>180	Wearable-only	0.907	0.59	0.48	90.2%	85 min	2.06
<70	Wearable-only	0.933	0.21	0.18	73.2%	95 min	0.51
>180	**Fused**	**0.899**	**0.56**	**0.47**	**87.7%**	**75 min**	**2.07**
<70	**Fused**	**0.933**	**0.21**	**0.17**	**68.3%**	**90 min**	**0.49**

First 24 hours (cold start)

Model	Event	AUC	AP
Wearable only	>180	0.875	0.49
Fused	>180	0.863	0.47
Wearable only	<70	0.804	0.06
Fused	<70	0.834	0.07

10,488 cold start moments per model.

Forecast and band on the sample profile

12.7 mg/dL error at 1 h (persistence 16.9)

17.5 mg/dL error at 2 h (persistence 25.1)

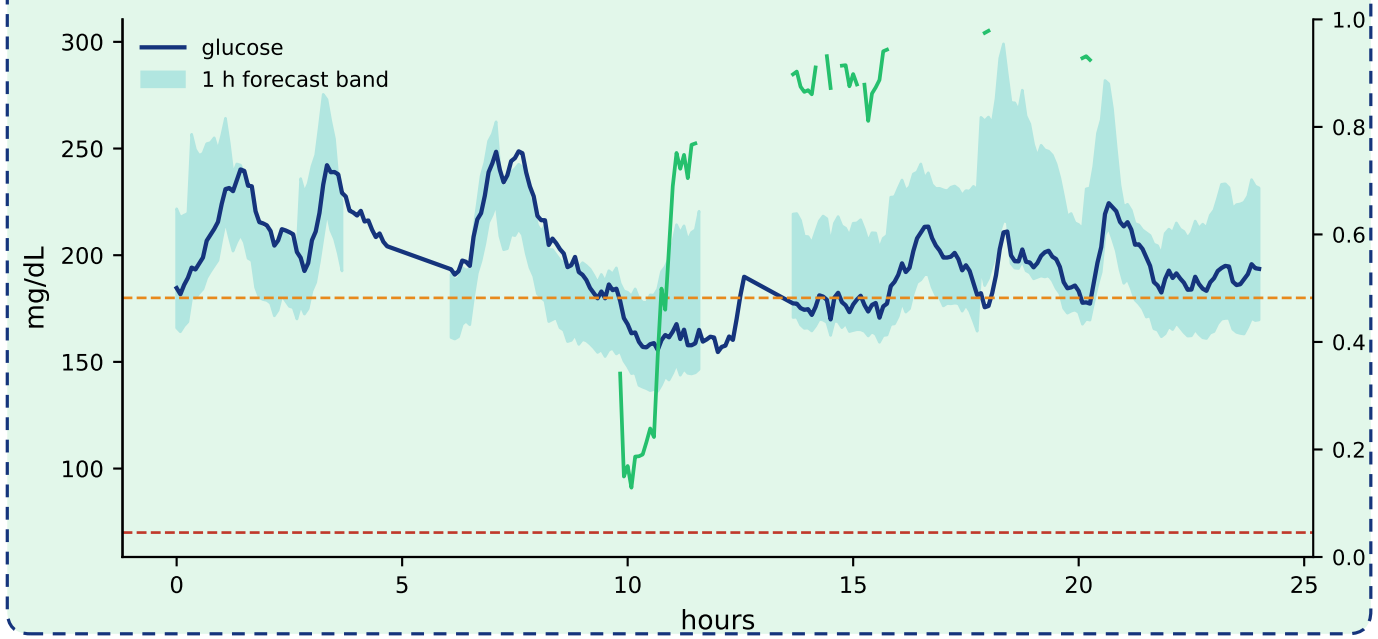
80% band coverage, widened 3.2 mg/dL at 1 h

The small cohort gives lower AUC than the 800 patient run; thresholds maximise F1 on pooled out of fold scores. Numbers here regenerate from data/sample/results.

Triage board (real sample output)

Patient	P(glucose above 180 within 2 h)
b6d46abe	97%
01ee9e96	97%
0832d9ad	95%
3fab2315	92%
d3727ff2	89%
227c0989	88%
3b24d024	87%
790ad9a1	86%

Patient view (real traces, patient 01ee9e96)



Why flagged

SHAP drivers from the fold model that never saw the patient, each bar tagged wearable or health record

What if

toggle a medicine or activity and the risk is recomputed; shown as association, not causal advice

Live monitor

replays a patient tick by tick and scores only the data received so far, with no look ahead

Alert log

each alert is marked confirmed or false once 2 hours have passed

Real time path: streaming equals batch

New readings arrive → Rebuild features from history so far → Score with held out fold model → Risk, band, SHAP → Alert needs 2 reads above threshold

tests::test_stream_features_equal_batch_and_scores_match_oof proves no look ahead.

Tests that guard the claims

Test	Checks
bmi_consistent, homa_ir_derived	derived fields are consistent
latest_labs_match, lab_visits_4_to_8	latest visit equals patient table
meds_match_conditions, complications_need_diabetes	clinical rules hold
genomics_complete_and_labeled	5 SNPs each, labelled SIMULATED
raw_schema_and_native_rates, small_gaps_only	native rates, only small gaps filled
fuse_no_future_lab_leakage	no future lab reaches a timestamp
event_label_uses_only_future_window, onsets_and_lead_time	label window and lead time
stream_features_equal_batch	live equals batch

Reproduce

- **sample** — 50 patients x 7 days, minutes
- **full** — 800 x 14 days, ~1 h on 10 cores
- **py 3.12** — Java 17 for Synthea, libomp

```
python run_pipeline.py --profile sample
streamlit run dashboard/app.py
```

A tiered design a clinic can run

Wearable and record data

800 patients, every 5 minutes

Model alerts patients first

about 2.4 alerts a day each at balanced

Triage board

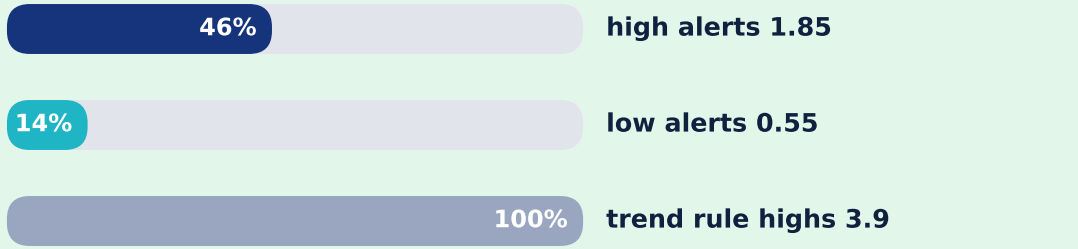
only patients in alert now

Clinician acts

who, when and why on one screen

The patient is alerted first; raw alert volume never reaches the doctor.

Alert volume per patient day





Relevance to India

Few clinicians serve many patients, and the risk sits between visits. A small team can supervise a large panel, and each clinic sets the sensitive, balanced or precise dial to its own staffing and risk tolerance.

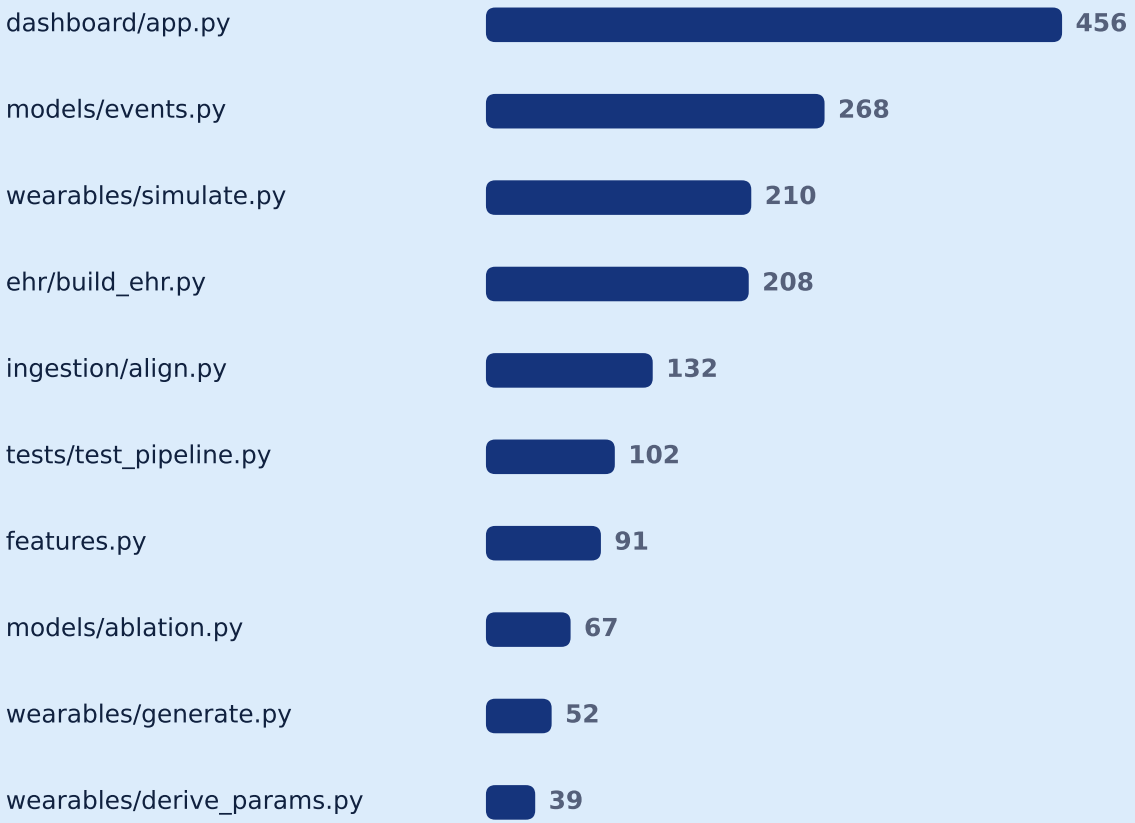
Limitations, stated plainly

1. **Simulator is EHR driven**
   fusion gain shows recovery of that link, not proof it exists in real patients
2. **Hypoglycaemia is weaker**
   about 65% of low alert episodes are false at balanced; unlogged insulin and meals remain
3. **Not a validated T2D model**
   type 1 UVA/Padova ODE with PI controller used as a generic glucose insulin system
4. **Optimistic thresholds**
   chosen on pooled held out scores; not tuned to a clinical cost
5. **Associations, not causes**
   what if answers are not treatment advice
6. **Simulated inputs**
   genotypes, polygenic score, devices; research code, not a medical device

Roadmap ahead

Validate	same ingestion adapter on consented real data: MIMIC IV for the record, an open CGM dataset for sensors
Calibrate	tune thresholds to a clinical cost of missed versus false alarms with clinicians
Connect	real device adapter for the live monitor, per patient online recalibration
Extend	reuse ingestion and fusion for hypertension or cardiovascular risk

Codebase at a glance (lines of Python)



About 1,600 lines in total. The simulator, ingestion, models, dashboard and tests are separate modules with a single run_pipeline.py entry point.

Pipeline stages and outputs

Stage	Output under data/<profile>/
synthea	synthea_raw (not committed)
ehr	ehr/*.csv, sim_params.csv
wearables	raw_wearables, truth (not committed)
ingest	timeseries.parquet, fused_timeline.parquet
models	results/ablation.csv and json
events	events.csv, json, event_risk.parquet, event_models

Run it yourself

```
uv venv --python 3.12 .venv
```

```
python run_pipeline.py --profile sample
```

```
python run_pipeline.py --profile full
```

```
streamlit run dashboard/app.py
```

```
pytest tests
```

Stack and licence

Layer	Tools
Synthetic EHR	Synthea (Apache 2.0, Java 17), custom generators
Wearables	simglucose (UVA/Padova ODE), behaviour model
Data	pandas, NumPy, SciPy, pyarrow
Modelling	LightGBM, scikit-learn, SHAP contributions
Interface	Streamlit, Plotly
Quality	pytest, 13 tests
Licence	MIT; dependencies keep their own

Deliverables

README	team, title, problem, stack, licence
LICENSE	MIT, public repository
docs/architecture.pdf	architecture diagram
docs/TeamTwins.pdf	this deck
Video	demo on YouTube, link in README

Same patient. Earlier warning.

Thank you from Team TeamTwins: Anisha Gupta (IGDTUW) and Vibhor Sharma (GL Bajaj)

